Threshold × k | Accuracy and teacher token share

Qwen3-1.7B student / Qwen3-14B teacher · 8 benchmarks × 25 configurations × 2 metrics = 400 values

MATH500

Baseline accuracy: Student 71.60% | Teacher 84.60%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	81.20	82.00	**83.80**	82.20	81.00
2.5	80.20	80.80	81.20	82.00	80.20
3	80.00	79.60	80.60	79.60	78.60
3.5	76.20	79.00	78.80	79.60	80.00
4	77.60	78.20	76.60	77.40	77.80

Color scale: 76.20 – 83.80

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	1.15	4.20	7.63	13.52	23.83
2.5	0.76	2.83	4.95	9.42	17.10
3	0.52	1.86	3.62	6.81	11.93
3.5	0.35	1.29	2.69	5.02	9.15
4	0.26	0.95	1.89	3.46	6.83

Color scale: 0.00 – 100.00

OlympiadBench

Baseline accuracy: Student 41.69% | Teacher 55.04%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	54.30	53.12	54.60	**55.49**	53.12
2.5	53.12	52.52	54.01	53.86	51.48
3	51.34	51.63	51.48	51.34	50.59
3.5	50.30	50.45	50.74	47.63	47.48
4	49.55	46.14	46.44	48.52	47.48

Color scale: 46.14 – 55.49

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	1.19	4.23	7.58	13.66	23.36
2.5	0.74	2.62	5.16	9.48	16.95
3	0.49	1.83	3.50	6.54	12.09
3.5	0.37	1.30	2.41	5.03	8.86
4	0.25	0.91	1.84	3.45	6.59

Color scale: 0.00 – 100.00

BBH

Baseline accuracy: Student 43.06% | Teacher 79.81%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	79.17	79.72	79.07	**80.00**	77.87
2.5	79.17	79.26	79.26	78.24	77.87
3	79.26	78.43	78.61	78.98	77.04
3.5	77.87	79.17	78.33	78.61	76.39
4	77.69	78.15	77.96	76.76	77.69

Color scale: 76.39 – 80.00

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	6.91	16.87	28.71	36.55	55.79
2.5	4.65	9.86	22.61	32.70	51.17
3	4.54	12.74	23.14	29.85	44.81
3.5	3.60	10.34	20.92	24.15	39.96
4	2.87	10.34	14.93	26.79	37.67

Color scale: 0.00 – 100.00

CRUXEval-O

Baseline accuracy: Student 21.88% | Teacher 46.75%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	45.50	45.12	44.88	45.00	45.38
2.5	45.50	45.12	45.00	45.12	45.25
3	45.88	45.75	45.12	45.50	45.62
3.5	46.00	45.88	45.38	45.50	45.75
4	**46.12**	46.00	45.25	45.50	45.75

Color scale: 44.88 – 46.12

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	2.41	6.01	12.48	13.85	13.40
2.5	2.36	4.96	9.33	11.08	10.46
3	1.57	7.04	8.06	9.09	10.46
3.5	1.45	5.96	6.20	7.96	8.88
4	1.35	4.83	5.66	8.48	8.43

Color scale: 0.00 – 100.00

CRUXEval-I

Baseline accuracy: Student 19.75% | Teacher 50.75%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	**49.00**	47.62	48.12	48.88	48.88
2.5	48.88	47.38	47.75	48.75	48.75
3	48.25	47.25	47.62	48.50	48.75
3.5	47.12	47.25	48.12	48.75	48.88
4	47.50	46.88	47.88	48.50	48.88

Color scale: 46.88 – 49.00

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	22.14	48.55	63.22	78.42	56.04
2.5	21.54	48.08	63.04	78.34	40.05
3	20.91	47.61	62.54	77.96	34.69
3.5	20.48	47.13	62.31	77.81	31.26
4	19.83	46.46	61.92	77.58	29.33

Color scale: 0.00 – 100.00

HumanEval+

Baseline accuracy: Student 62.80% | Teacher 87.20%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	81.10	81.71	82.93	**85.98**	84.15
2.5	82.32	82.32	82.32	85.37	84.15
3	80.49	79.88	82.93	84.76	82.32
3.5	82.32	79.27	82.32	83.54	83.54
4	84.76	79.88	82.32	82.32	81.10

Color scale: 79.27 – 85.98

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	3.58	11.41	19.18	29.10	42.57
2.5	3.18	10.63	18.50	28.00	41.26
3	3.09	10.08	17.71	26.71	39.99
3.5	2.86	9.58	16.73	25.45	39.58
4	2.62	8.96	15.60	24.46	37.02

Color scale: 0.00 – 100.00

MBPP+

Baseline accuracy: Student 51.85% | Teacher 72.49%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	71.96	72.49	73.02	72.22	72.22
2.5	73.28	72.75	72.75	72.75	71.96
3	71.96	72.75	71.69	72.22	71.96
3.5	73.02	**73.54**	72.22	73.28	72.49
4	73.28	72.22	72.75	73.02	72.75

Color scale: 71.69 – 73.54

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	6.68	19.16	30.76	43.63	58.01
2.5	6.17	17.81	28.88	41.72	55.95
3	5.49	16.94	27.14	39.76	54.14
3.5	5.12	16.18	25.66	37.86	51.48
4	4.80	15.25	24.42	36.70	50.05

Color scale: 0.00 – 100.00

LiveCodeBench

Baseline accuracy: Student 17.71% | Teacher 30.86%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	28.57	27.43	29.71	26.29	28.57
2.5	28.57	27.43	27.43	26.86	30.86
3	26.86	27.43	28.00	29.71	31.43
3.5	28.00	26.86	28.57	29.71	29.71
4	28.00	25.71	26.86	30.29	**32.00**

Color scale: 25.71 – 32.00

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	5.09	15.25	27.02	40.74	56.49
2.5	3.78	12.54	20.46	33.40	50.39
3	3.22	9.78	17.38	28.79	43.00
3.5	2.49	8.94	14.93	24.04	33.64
4	1.87	7.06	12.47	20.25	32.70

Color scale: 0.00 – 100.00

Accuracy, interventions and length: independently scaled per benchmark. Teacher share: shared 0–100% scale. Darker = larger. Bold = best hybrid accuracy.
Teacher share excludes prompts / discarded drafts; it is not compute cost. Stopping rules and baseline generation differ. Full source: source.md.

Threshold × k | Interventions and output length

Qwen3-1.7B student / Qwen3-14B teacher · 8 benchmarks × 25 configurations × 2 metrics = 400 values

MATH500

Baseline mean output tokens: Student 885.08 | Teacher 911.39; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	9.71	8.67	8.00	7.46	6.09
2.5	6.29	5.72	5.16	4.79	4.50
3	4.41	3.83	3.69	3.46	3.22
3.5	3.08	2.69	2.76	2.65	2.52
4	2.15	2.12	2.06	1.83	1.85

Color scale: 1.83 – 9.71

Mean output tokens | k →

Threshold	1	4	8	16	32
2	842.03	824.53	837.39	879.40	808.50
2.5	829.13	807.28	830.83	809.57	832.38
3	841.23	823.16	814.26	808.41	854.73
3.5	866.49	834.26	818.37	838.94	868.77
4	822.84	885.79	869.41	840.68	854.41

Color scale: 807.28 – 885.79

OlympiadBench

Baseline mean output tokens: Student 1604.45 | Teacher 1524.58; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	17.94	15.54	14.17	12.90	10.48
2.5	10.96	9.78	9.39	8.69	7.50
3	7.33	6.57	6.47	6.00	5.36
3.5	5.38	4.76	4.62	4.47	4.04
4	3.76	3.51	3.55	3.28	2.94

Color scale: 2.94 – 17.94

Mean output tokens | k →

Threshold	1	4	8	16	32
2	1510.33	1458.84	1482.34	1494.41	1409.54
2.5	1479.97	1482.64	1440.49	1441.71	1383.82
3	1482.96	1414.21	1456.42	1438.13	1377.94
3.5	1462.08	1443.61	1505.72	1387.82	1406.15
4	1523.21	1504.16	1515.12	1485.23	1379.40

Color scale: 1377.94 – 1523.21

BBH

Baseline mean output tokens: Student 479.75 | Teacher 279.43; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	18.61	13.09	10.25	7.49	5.24
2.5	15.78	11.37	9.01	6.79	4.86
3	13.80	10.05	8.12	6.22	4.56
3.5	12.12	9.07	9.34	5.66	4.25
4	10.94	8.15	6.74	5.27	4.03

Color scale: 4.03 – 18.61

Mean output tokens | k →

Threshold	1	4	8	16	32
2	269.08	303.92	274.39	306.16	268.25
2.5	339.22	451.22	304.97	306.82	268.37
3	304.23	307.83	267.48	306.75	284.82
3.5	336.78	341.02	342.84	341.50	292.69
4	380.88	305.89	341.00	284.98	292.08

Color scale: 267.48 – 451.22

CRUXEval-O

Baseline mean output tokens: Student 402.01 | Teacher 115.12; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	2.56	1.44	1.19	1.04	1.00
2.5	2.52	1.42	1.22	1.10	1.00
3	2.48	1.41	1.21	1.10	1.00
3.5	2.52	1.40	1.20	1.10	1.00
4	2.39	1.40	1.15	1.10	1.00

Color scale: 1.00 – 2.56

Mean output tokens | k →

Threshold	1	4	8	16	32
2	106.33	95.77	68.52	79.81	102.32
2.5	106.59	114.70	94.43	108.90	131.78
3	158.34	80.15	108.80	132.74	132.49
3.5	174.12	93.44	140.31	151.42	155.54
4	177.48	115.36	146.33	142.72	162.62

Color scale: 68.52 – 177.48

CRUXEval-I

Baseline mean output tokens: Student 349.06 | Teacher 18.23; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	4.07	2.21	1.61	1.21	1.02
2.5	3.95	2.19	1.60	1.21	1.01
3	3.82	2.17	1.59	1.21	1.01
3.5	3.73	2.15	1.59	1.21	1.01
4	3.62	2.13	1.58	1.21	1.01

Color scale: 1.01 – 4.07

Mean output tokens | k →

Threshold	1	4	8	16	32
2	18.36	18.16	18.20	18.14	27.58
2.5	18.35	18.24	18.17	18.14	38.53
3	18.28	18.22	18.21	18.21	44.45
3.5	18.23	18.21	18.20	18.18	49.26
4	18.25	18.30	18.27	18.21	52.49

Color scale: 18.14 – 52.49

HumanEval+

Baseline mean output tokens: Student 199.18 | Teacher 177.63; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	6.53	5.11	4.34	3.39	2.58
2.5	5.74	4.85	4.27	3.27	2.52
3	5.55	4.56	4.02	3.16	2.48
3.5	5.12	4.32	3.77	3.02	2.45
4	4.62	3.99	3.50	2.88	2.27

Color scale: 2.27 – 6.53

Mean output tokens | k →

Threshold	1	4	8	16	32
2	182.60	178.95	180.31	181.36	178.18
2.5	180.76	182.20	183.64	181.47	179.35
3	179.61	180.81	180.92	183.88	181.65
3.5	179.32	180.15	179.65	184.75	181.82
4	176.31	177.83	178.57	183.10	181.39

Color scale: 176.31 – 184.75

MBPP+

Baseline mean output tokens: Student 139.05 | Teacher 90.15; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	6.13	4.55	3.79	2.76	2.01
2.5	5.80	4.45	3.55	2.70	1.95
3	5.19	4.15	3.43	2.61	1.92
3.5	4.92	4.02	3.24	2.52	1.88
4	4.69	3.91	3.11	2.49	1.81

Color scale: 1.81 – 6.13

Mean output tokens | k →

Threshold	1	4	8	16	32
2	91.89	92.10	91.52	89.21	91.13
2.5	94.10	97.02	91.12	90.38	91.35
3	94.48	94.77	93.68	92.11	93.89
3.5	96.02	96.07	93.10	93.36	96.48
4	97.59	99.31	94.14	94.97	95.24

Color scale: 89.21 – 99.31

LiveCodeBench

Baseline mean output tokens: Student 618.22 | Teacher 546.82; interventions 0

Mean interventions | k →

Threshold	1	4	8	16	32
2	28.93	21.90	17.83	13.34	9.69
2.5	22.47	17.67	14.25	11.78	8.91
3	18.18	14.95	12.67	9.36	7.81
3.5	14.44	11.93	10.41	8.57	6.55
4	11.63	10.30	8.83	7.35	6.07

Color scale: 6.07 – 28.93

Mean output tokens | k →

Threshold	1	4	8	16	32
2	567.96	572.71	524.41	516.69	536.02
2.5	593.76	561.63	554.13	555.95	552.02
3	563.92	609.34	580.23	513.47	570.83
3.5	580.63	531.71	554.88	562.46	609.21
4	620.26	582.14	563.57	574.39	580.33

Color scale: 513.47 – 620.26

Accuracy, interventions and length: independently scaled per benchmark. Teacher share: shared 0–100% scale. Darker = larger. Bold = best hybrid accuracy.
Teacher share excludes prompts / discarded drafts; it is not compute cost. Stopping rules and baseline generation differ. Full source: source.md.